Active Inference – A Synthetic Overview

An applicable introduction across equilibrium and far-from-equilibrium systems

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1 Abstract

Active Inference casts perception and action as free-energy minimization. This overview shows how Bayesian mechanics carries the Free Energy Principle from equilibrium to far-from-equilibrium life.

2 Extended Abstract

A thing that persists must, in effect, hold a model of the world that keeps it intact. **Active Inference** turns that observation into mechanics: a system maintains itself by minimizing *variational free energy*, a single quantity that scores how badly its internal expectations fit incoming sensations [Friston, 2010, Parr et al., 2022]. Perception lowers free energy by updating beliefs; action lowers it by changing the world so the beliefs come true. One currency, two directions — inference and control — which is why the same template reappears from single cells to brains to institutions.

The bound is what makes this *applicable* rather than metaphorical. Variational free energy is an upper bound on **surprise**, the negative log evidence $-\ln p(s)$ a system assigns to its own sensory states. Because the bound is computable from a system's own beliefs, minimizing it *is* approximate Bayesian inference carried out by the physical dynamics of the thing itself. The accounting is organized by a **Markov blanket**: a statistical boundary splitting *external* states η from *internal* states μ, coupled only through *sensory* and *active* blanket states. Internal states come to parameterize beliefs about the external world they can never touch directly — the core claim of **Bayesian mechanics**, a physics written in terms of, and by, those beliefs [Da Costa et al., 2021, Ramstead et al., 2023].

fig. 1 shows the move that this overview is built around, without a single equation of derivation. Both panels live on *the same* free-energy landscape — a Gaussian bowl whose floor is the system's expected state. At **equilibrium** (panel a) the flow is pure descent, $f = -\nabla F$: every trajectory falls straight to the mode and stops. Detailed balance holds, there is no net current, and — taken literally — nothing further happens. A rock is at equilibrium; so, eventually, is a system that has stopped doing anything. Living systems occupy panel (b). There the flow gains a *solenoidal* term, $f = (Q - \Gamma)\nabla F$, that runs along the contours of F rather than down them. Free energy is still minimized — the system still tracks its niche — yet probability mass **circulates** indefinitely in a **non-equilibrium steady state**. That persistent current is the signature of things that stay far from equilibrium: heartbeats, foraging, perception that never settles. The gradient part is the Free Energy Principle's "climbing down surprise"; the solenoidal part is what keeps the system alive while doing so [Friston et al., 2023].

Read the two panels together and the headline of Active Inference is visible at a glance: the *same* imperative — minimize free energy — produces inert relaxation in one regime and open-ended, self-sustaining behavior in the other. The rest of this overview unpacks that picture in the spirit of *show, not tell*: each subsequent section adds one runnable, minimal demonstration rather than another layer of formalism.

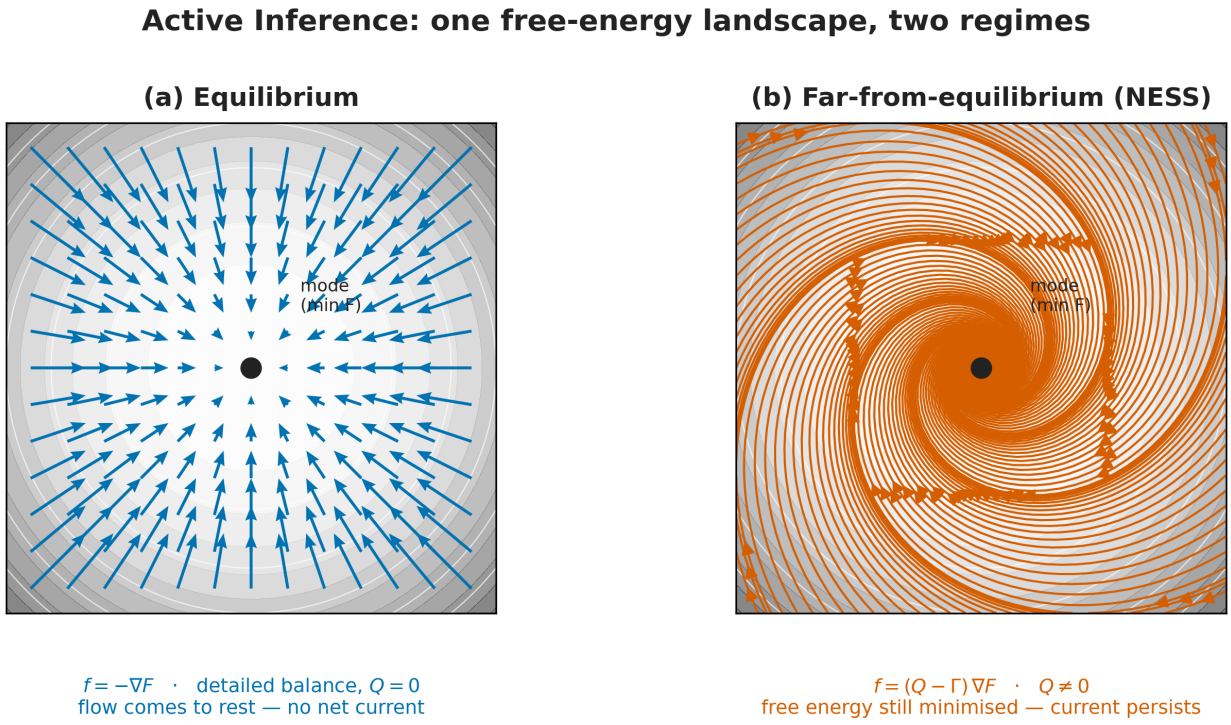


Figure 1: **One landscape, two regimes.** Both panels place the system on the same Gaussian free-energy bowl (grey contours; the black dot marks the mode, where free energy is least). **(a) Equilibrium:** the flow is pure gradient descent on free energy, $f = -\nabla F$ (blue arrows). Detailed balance holds, so trajectories fall to the mode and come to rest with no net probability current. **(b) Far-from-equilibrium (NESS):** an antisymmetric *solenoidal* operator Q adds a conservative current, $f = (Q - \Gamma)\nabla F$ (orange streamlines), that flows along the level sets of F. Free energy is still minimized, but probability mass circulates without end — the hallmark of living, self-organizing systems. The Free Energy Principle is the descent term; Bayesian mechanics adds the circulation. Internal states μ infer external states η across a Markov blanket (sensory s, active a). Figure rendered deterministically by `scripts/generate_overview_figure.py` over the tested flow in `src/free_energy_flow.py`.

3 References

Bibliography lives in `manuscript/references.bib` and is read by Pandoc during PDF render. The build pipeline invokes Pandoc with `--natbib`, so every `[@key]` citation in the manuscript is rewritten to the appropriate `\cite{}`/`\citep{}`/`\citet{}` LaTeX command and resolved against the bib file.

To validate that `references.bib` is syntactically clean and contains the required fields per entry type:

```
uv run python -m infrastructure.reference.citation.cli validate \
    projects/working/ActiveInferenceSynthetic/manuscript/references.bib --strict
```

References

Lancelot Da Costa, Karl Friston, Conor Heins, and Grigorios A. Pavliotis. Bayesian mechanics for stationary processes. *Proceedings of the Royal Society A*, 477(2256):20210518, 2021. doi: 10.1098/rspa.2021.0518.

Karl Friston. The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010. doi: 10.1038/nrn2787.

Karl Friston, Lancelot Da Costa, Noor Sajid, Conor Heins, Kai Ueltzhöffer, Grigorios A. Pavliotis, and Thomas Parr. The free energy principle made simpler but not too simple. *Physics Reports*, 1024:1–29, 2023. doi: 10.1016/j.physrep.2023.07.001.

Thomas Parr, Giovanni Pezzulo, and Karl J. Friston. *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press, Cambridge, MA, USA, 2022. ISBN 978-0-262-04535-3. doi: 10.7551/mitpress/12441.001.0001.

Maxwell J. D. Ramstead, Dalton A. R. Sakthivadivel, Conor Heins, Magnus Koudahl, Beren Millidge, Lancelot Da Costa, Brennan Klein, and Karl J. Friston. On bayesian mechanics: a physics of and by beliefs. *Interface Focus*, 13(3):20220029, 2023. doi: 10.1098/rsfs.2022.0029.